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Swarm intention identification via dynamic distribution probability image



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Abstract In the realm of decision-making for defense and security applications, it is paramount to swiftly and accurately identify the intentions of incoming swarms. Conventional identification methods predominantly focus on single-target applications and overlook the perturbations introduced by measurement noise. In this study, we propose a novel concept: the Dynamic Distribution Probability (DDP) image, which is constructed using the estimated state and its covariance matrix. Each grayscale pixel value within the image signifies the probability of the presence of the agent within the swarm. Our proposed identification scheme integrates the use of Extended Kalman Filter (EKF), Convolutional Neural Network (CNN), Back Propagation (BP) network, and Gated Recurrent Unit (GRU) network. Specifically, the DDP image is processed through a CNN to distill the formation characteristics, and the estimated swarm state from EKF is inputted into a BP network to deduce the kinematic information. The outputs from both networks are summed and subsequently channeled into a GRU network to capture temporal dynamics. Extensive numerical simulations and flight experiments are presented to demonstrate the robust anti-noise capability of the proposed scheme compared with conventional methods, as well as its superior training efficiency.

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1. Introduction

The advent of aerial swarm warfare has significantly transformed the contemporary battlefield landscape. This evolu-

tionary shift, has greatly magnified the intricacies of battle engagements and necessitates advanced strategies for enhancing quick situational awareness,^{1–5} intensifying the demands on commanders to require them to sift through vast volumes of battlefield data and extract critical information within constrained temporal windows.⁶ To bolster commanders' capabilities in comprehensively analyzing this influx of information and to promote optimal decision-making, the rapid and precise intention identification of incoming swarms becomes paramount. Given the considerable computational advantages computers possess over human cognitive processing speeds, they emerge as promising tools in this context. Thus, there is

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an urgent and significant impetus to design decision-support combat systems that can augment the efficiency of identifying the operations of incoming swarms.

In recent years, the artificial neural network has been utilized prolific applications in swarm operation and intention identification domains.^{7–12} Considering the complexity of intention identification problems, a single neural network is difficult to handle intricate environments. To address this issue Consequently, a hybrid neural network method, consisting of GRU network and BP network, for quick intention identification approach was presented in Ref. 13. The maneuver data is processed using a GRU network, while other data is inputted into a BP network. The authors in Ref. 14 employed the bidirectional GRU and condition random field with a space–time attention mechanism. More recently, the evidence network and belief decision trees are incorporated in Ref. 15. Although these approaches show effectiveness in single-aircraft scenarios, these methodologies encounter difficulties when applied to swarm identification. The study in Ref. 16 employed a tandem of the support vector machine and dynamic Bayesian network, alongside low-correlation features, to extract combat information and infer the intentions of multiple aircraft. Despite the inclusion of multiple aircraft in the scenario, the method primarily identifies the intentions of each individual agents in a sequential manner. Essentially, this approach emphasizes single intentions, positioning it fundamentally as a technique for single-target intention identification.

With the progression of warfare, swarm warfare is emerging as a novel trend in future battlefields. Current research, primarily targeting single-target intent, struggles to effectively handle the identification of swarm intentions. Although some methods designed for single aircraft claim to be capable of identifying the swarm intent via obtaining intentions of each agents, such methodologies tend to neglect the interrelations and interdependencies inherent among swarm agents. Indeed, the focal points of research in single-target and swarm intention identification diverge significantly. In the case of single-aircraft identification, the emphasis predominantly lies on the single-target attributes such as position, size, radar state, and payload. Conversely, when addressing swarms, the attention shifts towards swarm characteristics such as the formation dynamics and the inter-agent distances. Moreover, the single-aircraft intention identification methods often rely on some hard-to-obtain information (e.g., the status of radar system in Ref. 13 and electronic interference state in Ref. 14). These data requirements significantly diminish the practical applicability of these methods.

To address the shortcomings of research algorithms in the field of swarm intent identification, various countermeasure methods have been proposed.^{17–19} The authors in Ref. 20 introduce a deep learning architecture named W-CPCLSTM that consists of a contrastive predictive coding model, a variable-length long short-term memory network model, and an attention weight allocator for online intention recognition. In Ref. 21, an intention recognition model based on behavioral analysis and unsupervised learning clustering is proposed. This model uniquely identifies and groups the common intentions of multiple agents by analyzing the probability distributions over landmarks in their behavioral patterns. In Ref. 22, the dubins model is introduced to depict and study the dynamic characteristics of the movement of swarm. However, the aforementioned methods fail to integrate proactive measures for

noise resistance. In real-world scenarios, sensor measurements are rarely flawless and often contaminated by noise.^{23–26} Traditional approaches, being predominantly data-centric, exhibit a heightened sensitivity to data quality. Although noise can be artificially introduced into the training database to imitate real-world conditions, neural networks inherently like the capacity analyze noise in a targeted manner, and this operation potentially reduces training efficiency.¹³ Moreover, current researches lack a specific strategy for formations, which are in fact a crucial aspect affecting swarm intentions.

Motivated by aforementioned observations, we present a novel concept termed as the DDP image for swarm intension identification. Firstly, measurement associated with each agent are processed through EKF, yielding the estimated state and its associated covariance matrix. The DDP image is constructed based on the estimated state and its corresponding covariance matrix. Specifically, the grayscale values with in the image symbolize the probability of the presence of agents. This option provides an intuitive visualization of the noise covariance matrix from the filters, which can be swiftly assimilated by neural networks. We then propose an integrated identification scheme comprising CNN, BP network, and GRU network: the DDP image is channeled through a CNN to distill formation-centric data, while the swarm state is analyzed by a BP network to extract kinematic insights. The resultant outputs from these two networks are summed and inputted into a GRU network, enabling the capture of temporal dynamic information.

The remainder of this paper is organized as follows. The next section formulates the mathematics description of the intention identification problem. Section 3 introduces the concept of DDP image and structure of integrated identification scheme. The swarm intentions and database establishment method are defined in Section 4. The simulation results and flight experiment are presented in Section 5 and Section 6, respectively. The conclusions are given in Section 7.

2. Backgrounds and preliminaries

2.1. Mathematical models

We consider a scenario in which an incoming aerial swarm, composed of N agents, takes tactical actions against a military objective, as shown in Fig. 1, where xOy denotes the inertial frame and O is the observation location. The position and velocity of the i th agent are denoted by $[x_i, y_i]^T$ and

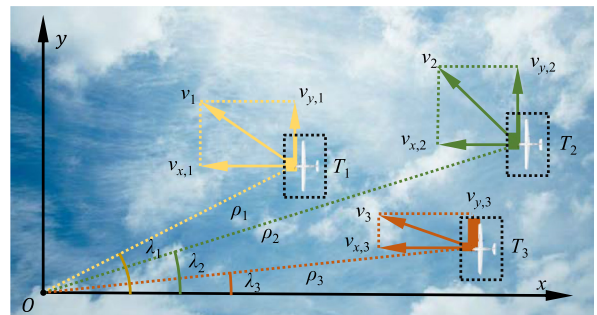


Fig. 1 Scenario of an incoming swarm.

$[v_{x,i}, v_{y,i}]^T$, respectively. The relative distance and bearing angle between the i th agent and the observer are represented by ρ_i and λ_i , respectively.

Define the state of each agent as $\mathbf{x}_i = [x_i, y_i, v_{x,i}, v_{y,i}]^T$ and the state of swarm as

$$\mathbf{x}_s = [x_s, y_s, v_{x,s}, v_{y,s}]^T = \frac{\sum_{i=1}^N \mathbf{x}_i}{N} \quad (1)$$

The relationship between the inertial position and the relative range as well as the bearing angle is determined by

$$\rho_i = \sqrt{x_i^2 + y_i^2} \quad (2)$$

$$\lambda_i = \arctan(v_{y,i}/v_{x,i}) \quad (3)$$

For the context of this study, we assume that the defending side possesses a radar system, facilitating the continuous monitoring of the relative range and bearing angle of each agent within the swarm. However, it is worth noting that in real-world scenarios, these measurements are contaminated by environmental noise, typically characterized as zero-mean white Gaussian noise. Therefore, the measurements can be expressed as $\mathbf{z}_i(t) = [\tilde{\rho}_i(t), \tilde{\lambda}_i(t)]^T$, where $\tilde{\rho}_i(t)$ and $\tilde{\lambda}_i(t)$ are noise-corrupted range and angle measurements, i.e.,

$$\mathbf{z}_i(t) = \begin{bmatrix} \tilde{\rho}_i(t) \\ \tilde{\lambda}_i(t) \end{bmatrix} = \begin{bmatrix} \rho_i(t) \\ \lambda_i(t) \end{bmatrix} + \mathbf{v} \quad (4)$$

where $\mathbf{v} \sim \mathcal{N}([0]_2, \mathbf{R})$ denotes the Gaussian measurement noise with $\mathbf{R} = \text{diag}(\sigma_\rho^2, \sigma_\lambda^2)$ being covariance matrix. The symbol σ_ρ and σ_λ stand for the standard deviations of the range and angle measurements, respectively.

2.2. Problem formulation

The operation of the swarm are intrinsically linked to its applied intentions. Identifying intentions can be conceptualized as a typical mapping problem, where one seeks to establish a correlation between the tactical intention and the observable information of the swarm:²⁷

$$\gamma(t_k) = f(\mathbf{z}(t_k)) \quad (5)$$

where $\gamma(t_k)$ is the applied tactical intention of swarm at time t_k and $\mathbf{z}(t_k) = [\mathbf{z}_1(t_k)^T, \mathbf{z}_2(t_k)^T, \dots, \mathbf{z}_N(t_k)^T]^T$.

Note that the realization of intent materializes through a sequence of actions. This indicates that using information from a single time instant as the input data for identification may yield intense fluctuations in outcomes. Hence, it is advisable to broaden the identification foundation from a single instant t_k to T -length time horizon as $[t_k - (T-1)\delta t, t_k - (T-2)\delta t, \dots, t_k]$, where δt denotes the sampling time. By using a sequence of measurements, the identification can be mathematically formulated as

$$\gamma(t_k) = f([\mathbf{z}(t_k - (T-1)\delta t), \mathbf{z}(t_k - (T-2)\delta t), \dots, \mathbf{z}(t_k)]) \quad (6)$$

Remark 1. The tactical intention is intricately linked to the battlefield condition, feature extraction processes, and other

environmental situations. This inherent complexity introduces a notable level of subjectivity and experiential dimension, rendering it challenging to establish a explicate mathematical representation. For this reason, this paper also suggests a learning-based approach to solve this problem.

3. Identification method

3.1. Overview of proposed approach

We propose an integrated identification approach comprising EKF, CNN, BP network, and GRU network. The structure of the proposed algorithm is shown in Fig. 2, where the representation of time is $t_1 \triangleq t_k - (T-1)\delta t, t_2 \triangleq t_k - (T-2)\delta t, \dots$. Firstly, the measurements are processed via certain data association technique. The measurements associated with each agent are then processed through corresponding EKF, yielding the estimated state $\hat{\mathbf{x}}_i$ and associated covariance matrix $\mathbf{P}_i$. Utilizing these results, we calculate the DDP image and the estimated swarm state at the current time instant. Subsequently, the DDP image is channeled through a CNN to distill formation-centric data, while the swarm state is analyzed by a BP network to extract kinematic information. The result from these two networks are summed and inputted into a GRU network to capture the temporal information. In the final stage, the softmax function is introduced in the output layer to calculate the probabilities of M intentions, i.e., $p_1(t_k), p_2(t_k), \dots, p_M(t_k)$.

Remark 2. The swarm target is a typical multi-target tracking problem. Considering the primary emphasis of this study is on the identification of tactical intentions within swarm behavior, we have intentionally simplified the tracking model in our approach. We assume that the number of measurements corresponds directly to the number of units in the swarm, and that the data association is accurately performed. Readers may refer to Refs. 28–32 for details of multi-objective tracking methods.

3.2. Building DDP image

For simplicity, in this subsection we denote $k \triangleq t_k$ and $k+1 \triangleq t_k + \delta t$. From the perspective of the defending size, the trajectories of incoming targets posses inherent uncertainty in real-world scenarios, necessitating the establishment of foundational assumptions regarding target maneuvers.³³ To optimize computational efficiency, particularly for onboard computing systems, we employ the linear constant velocity model as the foundation for our tracking estimator. The transition model is

$$\mathbf{x}_i(k+1) = \mathbf{\Psi}_i \mathbf{x}_i(k) + \mathbf{w}_i(k) \quad (7)$$

where $\mathbf{w}_i(k)$ is the process noise, which is assumed to be drawn from a zero mean multivariate normal distribution $\mathcal{N}$ with covariance $\mathbf{Q}$:

$$\mathbf{w}_i(k) \sim \mathcal{N}(0, \mathbf{Q}) \quad (8)$$

The transition model of constant velocity model can be expressed as:

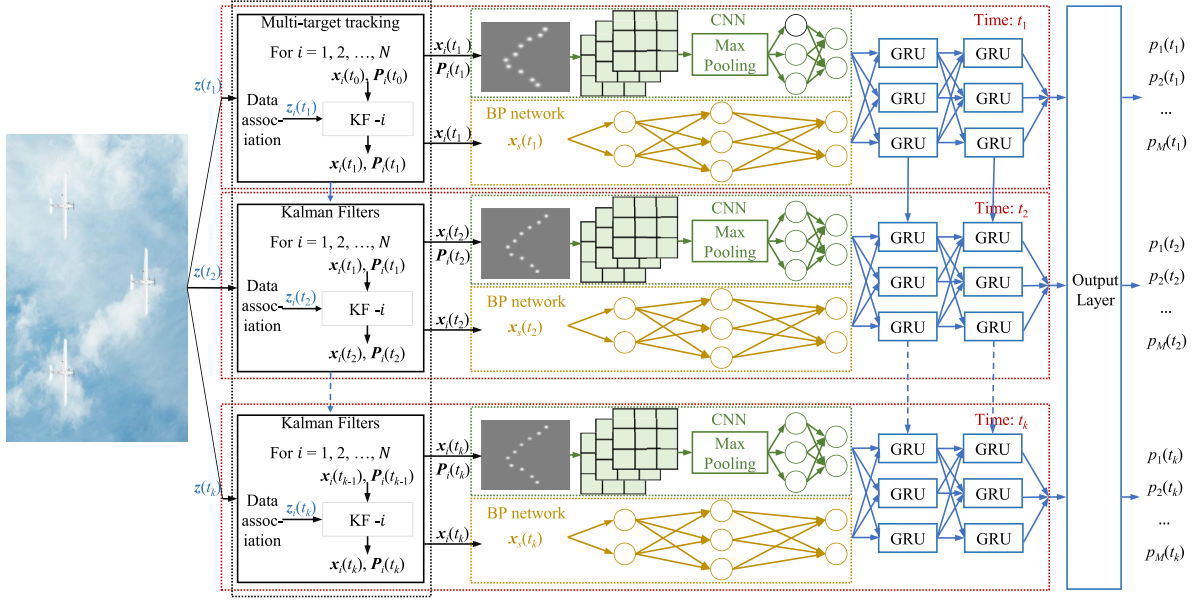


Fig. 2 Structure of identification model.

$$\Psi_i = \begin{bmatrix} 1 & 0 & \delta t & 0 \\ 0 & 1 & 0 & \delta t \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad (9)$$

According to Eqs. (2)–(4), the measurement matrix of the filter is

$$H_i = \begin{bmatrix} \frac{x_i}{\sqrt{x_i^2 + y_i^2}} & \frac{y_i}{\sqrt{x_i^2 + y_i^2}} & 0 & 0 \\ -\frac{y_i}{x_i^2 + y_i^2} & \frac{x_i}{x_i^2 + y_i^2} & 0 & 0 \end{bmatrix} \quad (10)$$

It is important to emphasize that while we use this specific model, other filter configurations and motion models can be seamlessly incorporated into the identification scheme proposed in this paper.^{34–37} The only requirement is the prompt computation of the state vector $\hat{x}_i$ and the covariance matrix P_i .

We build EKFs for each agent in the swarm. The input of the i th EKF is the corresponding measurement information $M_i(k)$, while the output is the estimated state $\hat{x}_i(k+1)$ and covariance matrix $P_i(k+1)$. The calculation process of EKF is

$$P_i(k+1|k) = \Psi_i P_i(k) \Psi_i^T + Q \quad (11)$$

$$S_i(k+1) = H(\hat{x}_i(k)) P_i(k+1|k) H(\hat{x}_i(k))^T + R \quad (12)$$

$$K_i(K+1) = P_i(k+1|k) H(\hat{x}_i(k))^T [S_i(k+1)]^{-1} \quad (13)$$

$$\hat{x}_i(k+1) = \hat{x}_i(k+1|k) + K_i(K+1) \times [M_i(k+1) - H(\hat{x}_i(k+1|k))] \quad (14)$$

$$P_i(k+1) = [I - K_i(k+1) H(\hat{x}_i(k+1|k))] P_i(k+1|k) \quad (15)$$

where $P_i(k+1|k)$ is the prediction error covariance matrix; $S_i(k+1)$ is the covariance of the innovations process; and $K_i(K+1)$ is the Kalman gain.

By using the estimated agent information from EKFs, the DDP image is derived from the estimated state $\hat{x}_i(k)$ and covariance matrix $P_i(k)$. This image effectively represents the probability distribution for each agent within the swarm. We

define the spatial separation between a specified point (x, y) and the estimated position of the i th agent as

$$e_i(x, y, k) = \begin{bmatrix} x - x_i(k) \\ y - y_i(k) \end{bmatrix} \quad (16)$$

According to the Gaussian distribution principles, probabilities drop off notably beyond triple standard deviation. Here, we consider the Mahalanobias distance defined as

$$D_{M,i}(x, y, k) = \sqrt{e_i^T(x, y, k) P_i^{-1}(k) e_i(x, y, k)} \quad (17)$$

The point is within triple standard deviation if $D_{M,i}(x, y, k) < 3$. Thus, the likelihood of the i th agent existing in (x, y) is expressed as

$$p_i(x, y, k) = \begin{cases} \frac{1}{2\pi\sqrt{|P_i(k)|}} e^{-\frac{e_i^T(x, y, k) P_i^{-1}(k) e_i(x, y, k)}{2}}, & D_{M,i}(x, y, k) < 3 \\ 0, & D_{M,i}(x, y, k) \geq 3 \end{cases} \quad (18)$$

The value $p_i(x, y, k)$ characterizes the spatial distribution likelihood for every agent within the swarm. A comprehensive distribution map of the swarm can be achieved by superimposing all individual agent maps. Given any point $[x, y]^T$ within the image, the probability of a particular agent being present is:

$$p(x, y, k) = 1 - \prod_{i=1}^N (1 - p_i(x, y, k)) \quad (19)$$

The subsequent phase involves transforming this spatial distribution into an image format compatible with a CNN, named DDP image. Since $p(x, y, k) \in [0, 1]$, we can apply a simple linear conversion to map the probability to grayscale values as

$$\tau(x, y, k) = p(x, y, k) \times 255 \quad (20)$$

When applying the proposed approach, we assume the swarm formation within a range of $x_i \in [x_s - \delta_x, x_s + \delta_x]$ and

$y_i \in [y_s - \delta_y, y_s + \delta_y]$, where δ_x and δ_y denote the pixel range variations in x and y axes, respectively. Given the potential variability in the formation of the incoming swarm and the unpredictability of its flight direction, we generally standardize the range by setting $\delta_x = \delta_y$. We then calculate the grayscale values of each image pixel point $\{(x, y) | x = x_s + k\delta_x, y = y_s + k\delta_y\}$ from Eqs. (16)–(20), where δ_x and δ_y are the resolution of the figure in x and y axes, respectively. Similar to analysis of δ_x and δ_y , generally we set $\delta_x = \delta_y$. The pseudo-code of building the DDP image is summarized in Algorithm 1.

Algorithm 1. DDP image formulation.

Data: $\hat{\mathbf{x}}_i(k), \mathbf{P}_i(k), i = 1, 2, \dots, N$
 Result: $\tau(x, y, k)$
 get the estimated swarm state $\hat{\mathbf{x}}_s$ by Eq. (1).
 for each pixel point (x, y) :
 • $i = 1, 2, \dots, N$:
 – calculate distance $e_i(x, y, k)$ by Eq. (16).
 – calculate probability $p_i(x, y, k)$ by Eq. (18).
 • calculate probability $p(x, y, k)$ by Eq. (19).
 • calculate grayscale value $\tau(x, y, k)$ by Eq. (20).

Remark 3. Notice that nonlinear transformations (e.g., an exponential transformation to accentuate central probability distributions) could also be potentially incorporated into this scheme to build the greyscale image. Our choice here is purposefully rooted in the computational efficiency. This ensures seamless compatibility with the processing capabilities of onboard computer systems.

3.3. Intention identification

We use the widely-accepted CNN to process the DDP image to extract formation characteristics and leverage the standard BP network to analyze kinematic property from the swarm state. The outputs from these two networks are summed and fed into a GRU network, facilitating the capture of temporal information. In the output layer, the softmax function is utilized to compute the probabilities associated with each intention.

The CNN consists of three layers: a convolution layer, a pooling layer and a fully connected layer. The convolutional layer receives the DDP image as its input and extracts features from the image by using a set of convolutional filters, which, unlike fully connected layers, establish sparse connections to scan local regions of the image. Following the convolutional layer, a pooling layer aims to down-sample the feature maps while retaining crucial information. In this paper, a max-pooling operation is used, which involves partitioning the feature map into non-overlapping regions and selecting the maximum value from each region. Post the pooling operation, the reduced feature maps are flattened into a one-dimensional vector, which is then fed into a fully connected layer.

Parallel to the CNN, we introduce a three-layer BP network to process the swarm state. We select the ReLU activation function in the BP network for its computational efficiency and simplicity. Nevertheless, it is worth noting that other

activation functions, such as Sigmoid and tanh, are fully compatible with the proposed structure. Considering the diverse scope of these input attributes, it is vital to normalize the input data. This normalization helps to ensure that attributes with larger spans do not overshadow those with tighter ranges during network training.^{38,39} Here we employ the commonly used maximum-minimum method for normalization as

$$x'_s(j) = \frac{x_s(j) - \min(x_s(j))}{\max(x_s(j)) - \min(x_s(j))}, \quad j = 1, 2, 3, 4 \quad (21)$$

where $\min(x_s(j))$ and $\max(x_s(j))$ are the minimum and maximum value of the j th element of vector $\mathbf{x}_s$.

Subsequently, we build a GRU network composed of four layers: an input layer, two GRU hidden layers, and an output layer. The inputs are derived from the summation of the CNN and BP outputs. In the hidden layer, the temporal information is extracted by basic GRU neurons. A basic GRU neuron consists of four parts: a reset gate $\mathbf{r}(k)$, an update gate $\mathbf{n}(k)$, a candidate state $\tilde{\mathbf{h}}(k)$ and an output state $\mathbf{h}(k)$. The algebra manipulations of GRU neurons are as determined by

$$\mathbf{r}(k) = \text{sigmoid}(\mathbf{w}_{hr}\mathbf{h}(k-1) + \mathbf{w}_{xr}\mathbf{x}^G(k)) \quad (22)$$

$$\mathbf{n}(k) = \text{sigmoid}(\mathbf{w}_{hn}\mathbf{h}(k-1) + \mathbf{w}_{xn}\mathbf{x}^G(k)) \quad (23)$$

$$\tilde{\mathbf{h}}(k) = \tanh(\mathbf{w}_{h\tilde{h}}\mathbf{r}(k) \times \mathbf{h}(k-1) + \mathbf{w}_{x\tilde{h}}\mathbf{x}^G(k)) \quad (24)$$

$$\mathbf{h}(k) = (1 - \mathbf{n}(k)) \times \mathbf{h}(k-1) + \mathbf{n}(k) \times \tilde{\mathbf{h}}(k) \quad (25)$$

where $\mathbf{w}$ is the weight matrix and $\times$ is the element-wise multiplication. Note that in a GRU neural network, the input data $\mathbf{x}^G(k)$ is equal to the output state $\mathbf{h}(k)$ of the previous layer.

In the output layer, the probabilities of each intention are calculated using softmax function as

$$\begin{aligned} \mathbf{p}(k) &= [p_1(k), p_2(k), \dots, p_M(k)]^T \\ &= \text{softmax}(\mathbf{w}_{\text{out}}\mathbf{h}^G(k) + \mathbf{b}_{\text{out}}) \end{aligned} \quad (26)$$

where $\mathbf{h}^G(k)$ is the output state of GRU neurons in the last GRU hidden layer.

Remark 4. In our research focused on intent recognition, the well-known challenge of GRUs in capturing long-term dependencies is not a major concern. This is because our field commonly deals with shorter input sequences, typically not exceeding 50 steps, which aligns with our study's use of 30-step sequences at a 10 Hz frequency. Hence, the standard GRU architecture suffices without modifications for our specific application.

4. Swarm intention definition and database establishment

We consider a heterogeneous swarm consisting of three prevalent types of aircraft:

(1) Search agent: An aircraft equipped with combat sensors (e.g., a radar or an electro-optical pod).

(2) Defense agent: An aircraft carrying small attack payloads (e.g., continuous rod warhead, small fragmenting warhead) and therefore having a higher maneuverability, whose response includes expelling interception and safeguarding other agents within the swarm.

(3) Attack agent: An aircraft with substantial attack payloads (e.g., incendiary warhead, large fragmenting

warhead) and therefore having limited maneuverability ability. The primary objective of this type of agent is to attack the high-value target.

A detailed overview of their individual characteristics are shown in Table 1.

By referring to single aircraft intention,^{27,40,6,13–16} we summarize seven typical swarm tactical intentions in this paper as:

(1) Penetration: The primary goal of the swarm is to break through the defensive line of the adversary, paving way for subsequent forces or facilitates strikes on vital targets beyond these defenses. Refer to Fig. 3(a) for a characteristic swarm formation with penetration intention.

(2) Detour: The swarm aims to circumvent the opposing defensive lines, inducing the adversary to reposition away from their optimal defensive stance and increasing the choice of successful subsequent offensives. See Fig. 3(b) for a typical formation illustrating a detour intention.

(3) Reconnaissance: The swarm conducts extensive surveillance of a region. Fig. 3(c) demonstrates a typical swarm formation for reconnaissance intention.

(4) Search & strike: The swarm composed of search and attack agents scours a designated area, identifying and attacking potential targets. Refer to Fig. 3(d) for a representative formation of this intention.

(5) Fast pass: The swiftly passes through a specific area, generally maintaining a dense formation. For a formation of fast pass, see Fig. 3(e) for example.

(6) Sequential attack: The swarm systematically attack a target at predetermined time intervals. A typical formation for this intention is shown in Fig. 3(f).

(7) Salvo attack: The swarm's objective is to attack a target with synchronized hits, enhancing the successful strike probability. Fig. 3(g) depicts a common formation for a salvo attack.

The attributes of each intention are shown in Table 2. In the column 'agent sort', the sequence of agents indicates the likelihood of utilizing a particular agent type of the corresponding intention.

Consider there is no open database for swarm intentions, we build the training and testing database through numerical simulations within a virtual combat simulation system. We employ the Latin hypercube sampling technique to set the applied intention and intention parameters for each simulation. From the obtained trajectory clusters, we randomly extract 10% trajectories for the test database, ensuring representation from each intention. The remaining trajectories are designated for the training database. Consequently, while both the testing and training database are constructed using the same approach, they remain mutually exclusive with no overlaps.

We then use the sliding windows to extract samples, as illustrated in Fig. 4. The input data includes the available swarm information in Eq. (4), while labels are constructed using the one-hot encoding method. Within the training dataset, all sample lengths are standardized to a predetermined length,

facilitating efficient network training. Kindly note that the interval for sliding windows does not necessarily match with the simulation time interval. For instance, while the EKF might have a measurement interval of 0.1 s, the sample extraction could occur every 0.5 s. This is because that over-sampling from a particular trajectory can lead to data redundancy after reaching an optimal generalization capability, which in turn, prolongs the simulation time and demands significant computational resources.

Remark 5. Note that in practice, besides including as many maneuvers as possible in the training dataset, we can customize intention types or modify intention parameters based on the actual combat environment. To accommodate these modified intentions, transfer learning^{41–44} can be effectively employed.

5. Simulation results

In this section, we validate the effectiveness of the proposed intention identification scheme through numerical simulations. We start by detailing the training parameters and processes. Following that, we present sample runs to offer an intuitive depiction of the identification performance. The robustness against image resolution and measurement noise is evaluated using Monte Carlo simulations.

5.1. Training process

The experimental samples are extracted using the database establishing method presented in Section 4. The measurement rate of combat sensors is 10 Hz, with a standard noise deviation $\sigma_p = 20$ m and $\sigma_\lambda = 0.1^\circ$, respectively. The covariance matrix in Eq. (8) is properly tuning as $\text{diag}(1, 1, 1, 1)$. The neural networks are constructed using the TensorFlow-1.13.0 framework. The structure parameters of networks and DDP image is shown in Table 3.

A total of 34886 samples are extracted, of which 31398 are allocated for training and the remaining 3488 for testing the model's generalization capabilities. To counteract overfitting, a dropout mechanism is embedded in the network, maintaining a dropout rate of 50%. We employ the Adam optimization algorithm, which leverages momentum and facilitates adaptive parameter training.⁴⁵ The batch size per iteration is set as 64 with a learning rate of 0.002, decaying at a rate of 0.99 per 100 iterations. The hyperparameters of the Adam optimizer are selected as $\epsilon = 10^{-8}$, $\eta_1 = 0.9$, and $\eta_2 = 0.999$.

Fig. 5 shows the accuracy and loss of the network during training process. As evident from the figure, after approximately 20 epochs, the accuracy surpasses 90%. After that, the loss steadily approaches convergence while the accuracy stabilizes around 95%. This trend signifies the completion of the model's training process.

5.2. Identification results

To provide an intuitive demonstration of the trained model's performance, we present two representative sample runs in this subsection. The performance of the proposed method is compared with the conventional model (named W-CPCLSTM in Ref. 20). The trajectory of a swarm with search and attack

Table 1 Kinematics characteristics.

Agent type	Speed (m/s)	Maximum maneuver (m/s ²)
Search agent	[30, 80]	25
Defense agent	[60, 100]	40
Attack agent	[60, 90]	20

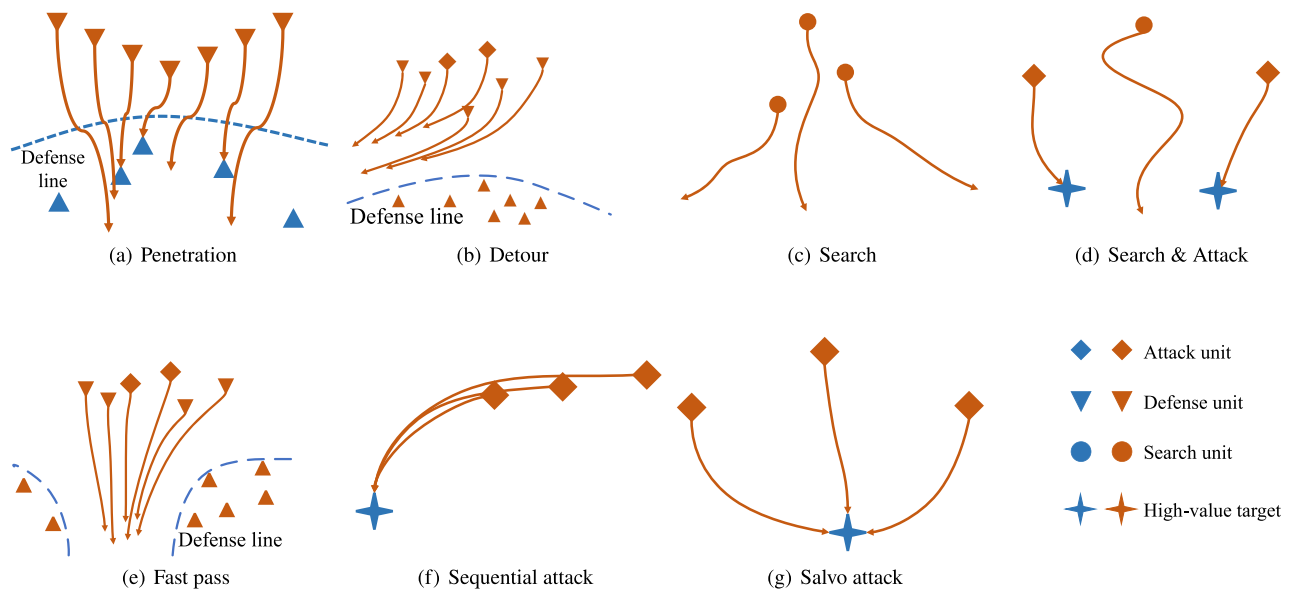


Fig. 3 Typical intention formation.

intentions is depicted in Fig. 6. In this scenario, a search agent conducts random surveillance maneuvers, while three attack agents fly in a straight line to attack potential targets. Specifically, for the exposed target located at the origin, the lowest attack agent is dispatched to carry out the attack. The transition processes of the proposed scheme and GRU-LSTM model are illustrated in Fig. 7. As can be seen from the figure, at $t = 0$ s, both models start with equal probabilities for each intention. Then the proposed model exhibits a rapid convergence, with the probability of identifying the true intention surpassing 50% within 1 s. In contrast, the model takes significantly longer, reaching the same probability threshold after about 1.9 s.

A more complex combat scenario is presented in Fig. 8. An incoming swarm consisting of six agents intends to attack the target located at (0, 0) m. The swarm initially performs a dense

formation with the fast pass intention. At $t = 10$ s, the swarm expands its formation and transits to the penetration intention. At $t = 35$ s, we assume that the swarm finds the frontal defense force is too strong and hence it shifts to the detour intention. At $t = 75$ s, the swarm once again expands its formation and executes the cooperative attack intention. At $t = 114.6$ s, the agents hit the target simultaneously.

The transition process of the proposed scheme and conventional network are shown in Fig. 9 and Fig. 10, respectively. The presented figures clearly demonstrate that the proposed scheme excels in rapidly and accurately identifying true intentions. This contrasts sharply with the performance of the conventional network, which initially errs in its identification (as seen in the first stage, fast pass intention). This discrepancy may be attributed to the larger initial distance of the swarm, which amplifies measurement noise and consequently broadens the swarm's perceived distribution range from a defense standpoint. Such an effect can erroneously present the appearance of a penetration intention, leading to incorrect intention identification by the conventional network. Moreover, when examining the transition dynamics of changing intentions, the proposed scheme consistently discerns shifts in intention within a time frame of 3 to 10 s. In contrast, the conventional network lags significantly in identifying these changes, exhibiting both slower response times and greater fluctuations. This comparison underscores the proposed scheme's effectiveness in complex scenarios, where it reliably identifies the intentions of an incoming swarm. Additionally, the proposed scheme demonstrates consistent and stable possibility lines in accurately identifying intentions, unlike the conventional model which shows more pronounced oscillations in the likelihoods of true intentions in subsequent simulations. This points to a more dependable and consistent performance from the proposed scheme.

We then use the testing dataset to test the performance for each intention. The intention identification accuracy of the proposed model and the conventional model is presented in Table 4. As can be seen from the table, the proposed provides

Table 2 Swarm intention attributes.

Intention	Number of agent	Distance between agents (m)	Agent sort
Penetration	[3, 12]	[100, 150]	defense agent attack agent
Detour	[5, 11]	[40, 200] m	search agent attack agent defense agent
Reconnaissance	[2, 3]	[100, 500]	search agent
Search & strike	[3, 7]	[100, 300]	search agent attack agent defense agent
Fast pass	[3, 10]	[40, 80]	defense agent defense agent attack agent
Sequential attack	[2, 6]	[100, 500]	search agent attack agent defense agent
Salvo attack	[2, 6]	[100, 500]	attack agent defense agent

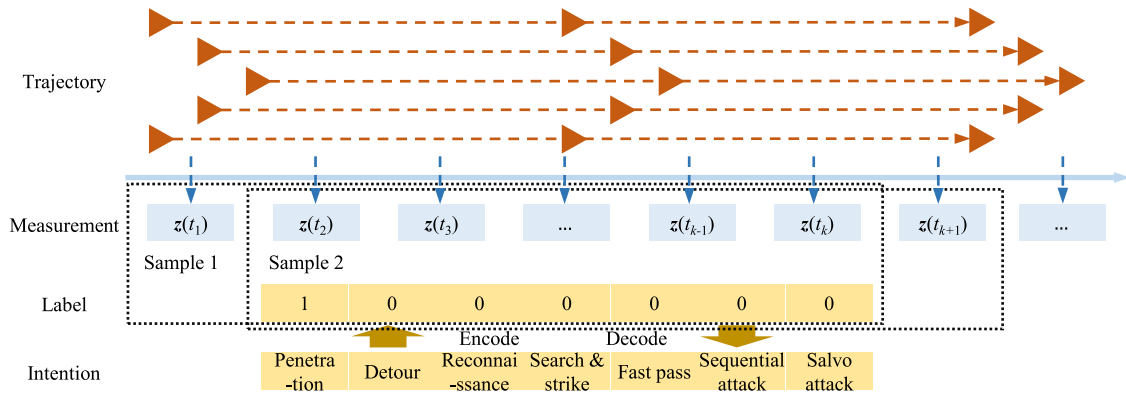


Fig. 4 Sliding windows for extracting samples.

Table 3 Neural network structural parameters.		
Component	Parameter	Value
CNN	Filter size	3×3
	Number of filters	2
	Pooling size	3×3
	Pooling stride	2
BP Network	Neurons (output layer)	96
	Neurons (input layers)	64
	Neurons (hidden layers)	64
	Neurons (output layer)	96
GRU Network	Neurons (input layer)	96
	Neurons (hidden layers)	64
	Step size K	30
	Resolution (δ_x, δ_y) (m)	10
DDP image	Resolution (δ_x, δ_y) (m)	10
	Range (δ_x, δ_y) (m)	500

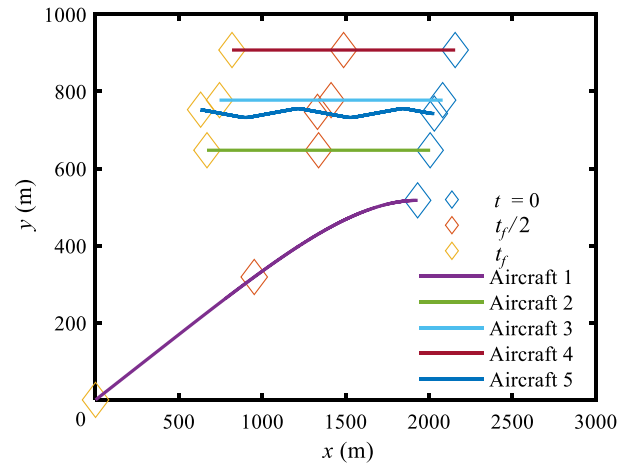
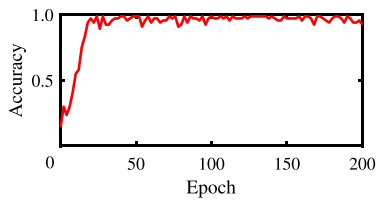
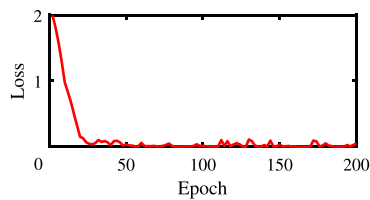


Fig. 6 Trajectory of sample run.



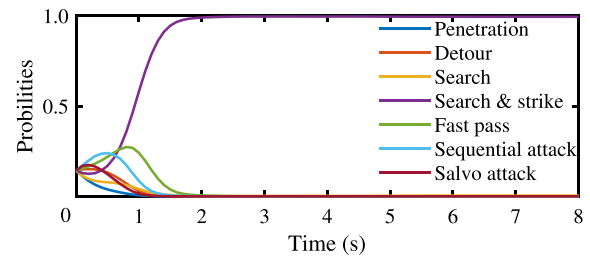
(a) Accuracy



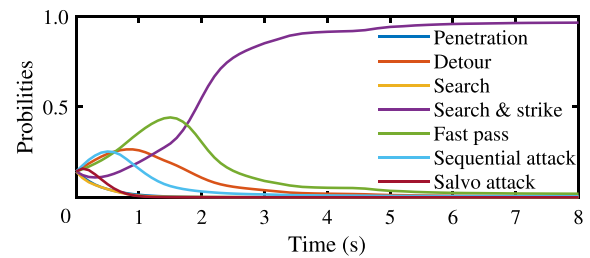
(b) Loss

Fig. 5 Training process.

better identification performance in all intention situations, validating the advantage over the conventional model. A crucial consideration is the real-time applicability of our model. During the simulation, the mean step computation time



(a) Proposed method



(b) Conventional method

Fig. 7 Transition processes.

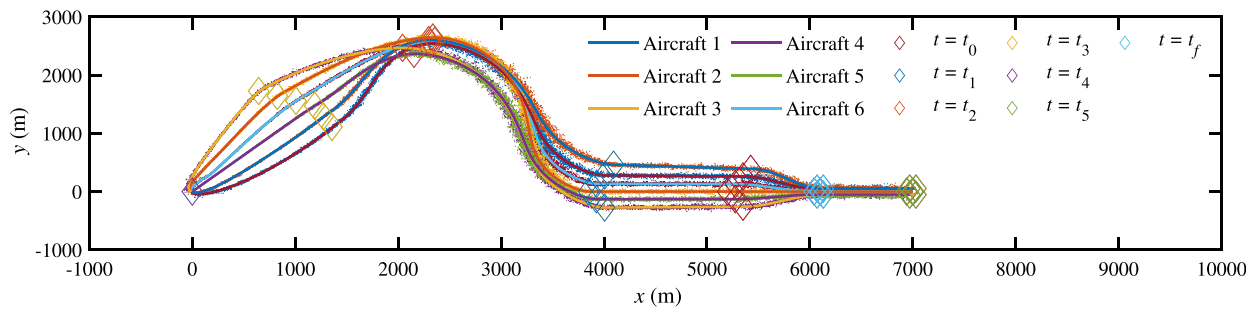


Fig. 8 Trajectories of situation.

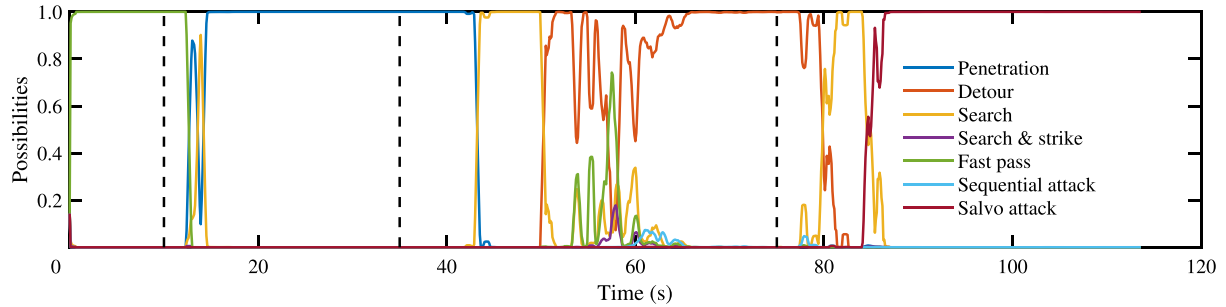


Fig. 9 Transition process of proposed method.

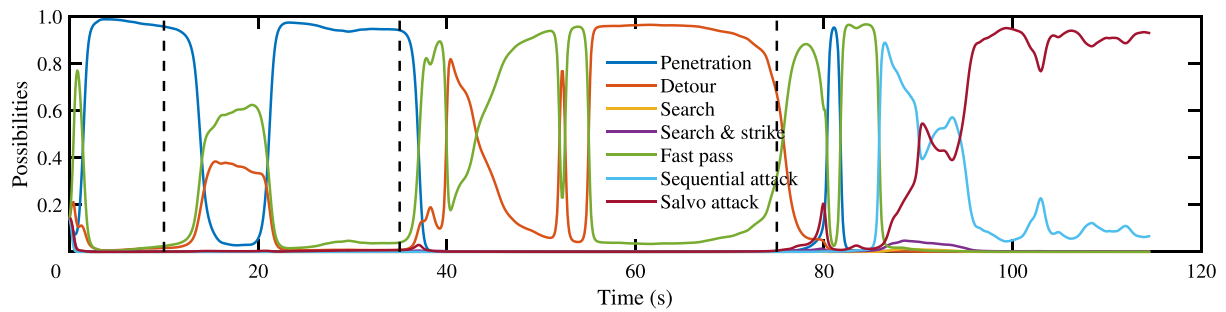


Fig. 10 Transition process of conventional network.

clocked at 13.046 ms. With a measurement interval of 100 ms, which considerably exceeds the computational time, we can conclude that the proposed method can achieve real-time computation requirement, making it a potential choice for practical applications.

5.3. Ablation experiment

In this subsection, we present the results of an ablation study designed to evaluate the contribution of individual components of our proposed model to the intent recognition task. The results of this experiment are summarized in Table 5. In the table, ‘formation’ refers to the swarm formations processed by the CNN, ‘kinematics’ denotes the kinematic information of the swarm processed by the BP neural network, and ‘dynamics’ represents the temporal information integrated by the GRU network. The findings reveal that the exclusion of any single component, be it formation or kinematics, leads

Table 4 Identification accuracy for each intention.

Intention	Accuracy	
	Proposed model	W-CPCLSTM model
Penetration	0.9543	0.9133
Detour	0.9347	0.9233
Search	0.9959	0.9714
Search & strike	0.9657	0.9778
Fast pass	0.9469	0.9143
Sequential attack	0.9171	0.8987
Cooperative attack	0.9388	0.9343
Mean	0.9505	0.9333

to a considerable decline in recognition accuracy. This decline underscores the importance of each component in capturing comprehensive information about the swarm. The results

demonstrate that omitting any single feature (formation or kinematics) leads to a significant decrease in recognition accuracy. The CNN-BP configuration, while lacking dynamic information, maintains a higher accuracy compared to CNN-GRU and BP-GRU. This suggests that having complete information at a single time point is valuable, but still less effective than the full model configuration (CNN-BP-GRU), which integrates all features over time for improved accuracy.

5.4. Robustness of proposed approach

We first investigate the influence of image resolution δ_x on the performance of the proposed model is explored in this subsection. We introduce the ‘successful-training rate’ as a metric, representing the percentage of models that achieve an identification accuracy exceeding 90% post-training. This rate offers a reliable measure to assess the effectiveness of the training process. A superior success rate indicates a stronger propensity for the model to identify an optimal gradient descent path during its training. It is important to underscore that all training sessions are conducted under consistent conditions. The rate of proposed schemes with $\delta_x = 5$ m, 10 m, 20 m, 50 m, and 100 m are presented in Fig. 11. We can find that the ratio reaches 100% when δ_x is less to 20 m. Yet, when the resolution expands to 50 m or more, the challenge of effectively training the model intensifies. This trend can be attributed to the fact that bigger δ_x generates less useful point in the DDP image, as described in Eq. (18). And the image abundant in useful points conveys richer information, thereby producing a clearer representation of the noise distribution.

Given the complexity of practical scenarios, the intention database might need to be frequently expanded and modified, which leads to frequent model training and retraining. A high success rate enables us to quickly determine whether the current training parameters are conducive for more in-depth training. Consequently, this creates an opportunity to allocate more time for deeper training. Nonetheless, while higher resolution increases the rate, it also introduces more computational burden and hence lengthens the training time. Thus, the optimal resolution should be chosen based on the specific needs and limitations of the particular scenario.

Now, let us explore the influence of noise on the performance of proposed models with different ratio. As depicted in Fig. 12, the robustness ability of the model notably decrease when δ_x reaches 20 m. This decline can be explained by the decrease of the useful pixel points in the image, which renders it inadequate in capturing the characteristics of positional distribution. In contrast, when resolutions δ_x are set as 5 m and 10 m, the model exhibits remarkable resilience to noise. Even when confronted with a measurement error as high as 50 m, it sustains an identification accuracy exceeding 90%. This

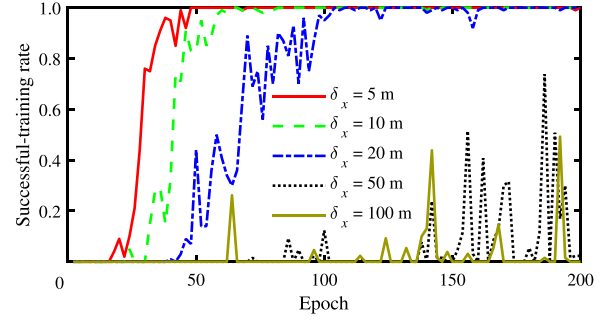


Fig. 11 Successful-training rate of different models.

validates that with sufficiently high resolution, the proposed model is well-equipped to eliminate the effects of noise.

6. Flight experiment

In this subsection, we seek to validate the robustness and accuracy of the proposed method by undertaking flight experiments. As shown in Fig. 13, the aircraft are equipped with a GPS and a camera, which can be used to record the geographic location data and flight environment. We first establish swarm intention trajectories in the inertial frame. Given the variance in scene sizes, we established a scaling ratio that varied from 10 to 30. Subsequent to the scaling, this data was converted into geographic longitude and latitude coordinates, as shown in Fig. 14. The swarm flies according to corresponding trajectories, while the sample are subsequently extracted and input into our pre-trained neural network based on the GPS recorded position information. Kindly note that the coordinate should be transformed from $[x_i, y_i]$ into $[\rho_i, \lambda_i]^T$ using Eqs. (3) and (2).

We presented two scenarios in the flight tests. In the first scenario, a swarm composed of six aircraft conducts a detour intention. In the second scenario, a swarm composed of seven aircraft conduct the penetration-detour-salvo-attack intention combination. The image captured on-site during these flight tests and the extracted trajectories used for intention identification are presented in Fig. 15 and Fig. 16, respectively.

The transition process of two scenarios are presented in Fig. 17 and Fig. 18, respectively. From the figure, it is noteworthy that conventional models exhibit poor performance in real flight tests. It requires approximately 3.5 s to correctly identify intentions even in simple scenarios, and display significant fluctuations in recognition results in composite scenarios. This demonstrates that their practical application potential is

Table 5 Ablation study results.

Network	Characteristic	Accuracy
CNN-BP-GRU	formation, kinematics, dynamics	0.9505
CNN-GRU	formation, dynamics	0.6251
BP-GRU	kinematics, dynamics	0.5732
CNN-BP	formation, kinematics	0.8673

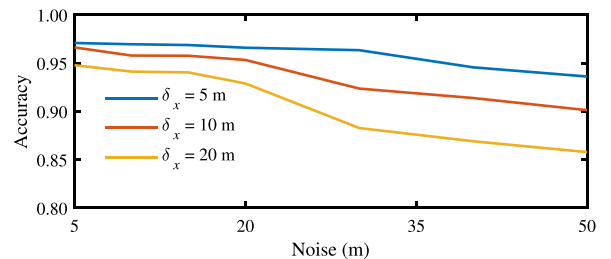


Fig. 12 Influence of noise.

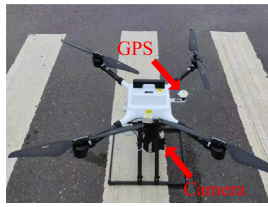
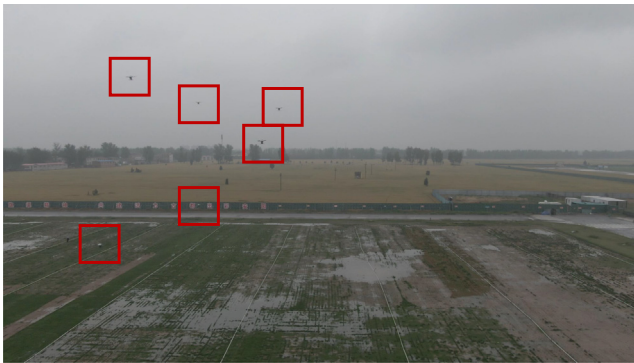


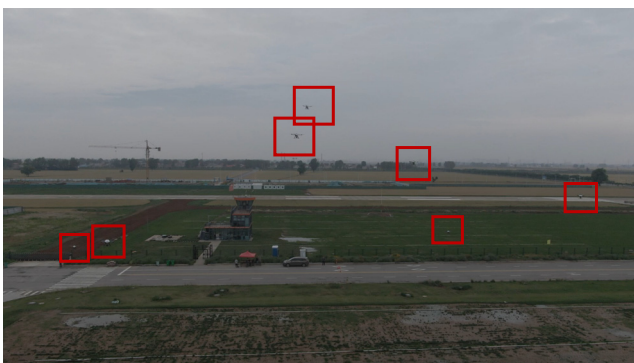
Fig. 13 Aircraft and record system.



Fig. 14 Experimental coordinate.



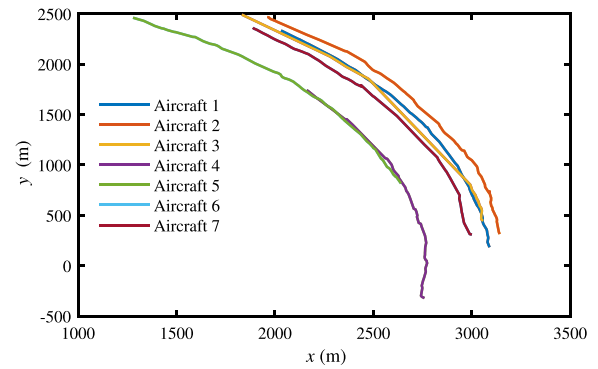
(a) Detour intention



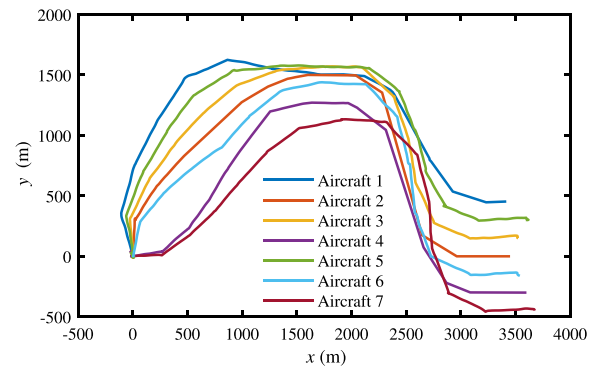
(b) Combination intention

Fig. 15 On-site figure.

significantly lower compared to the proposed solution. Meanwhile, we can find that the identification process depicted in flight experiment reveals pronounced fluctuations contrasted with performance during mathematical simulations in Fig. 7. These fluctuations underline the differences between idealized,

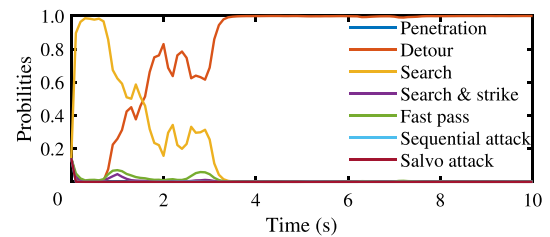


(a) Detour intention

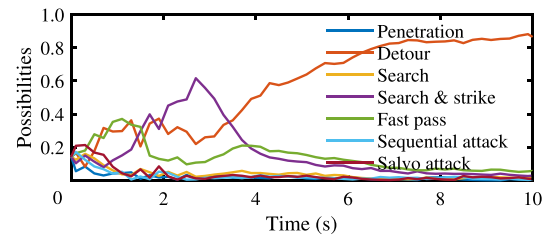


(b) Combination intention

Fig. 16 Flight trajectories.



(a) Proposed model



(b) Conventional model

Fig. 17 Transition process of Scenario 1.

controlled simulations and real-world flight conditions. The main difference comes down to unexpected noise in the measurement data. Although we deliberately added a predictable kind of noise in mathematical simulations to mimic real-

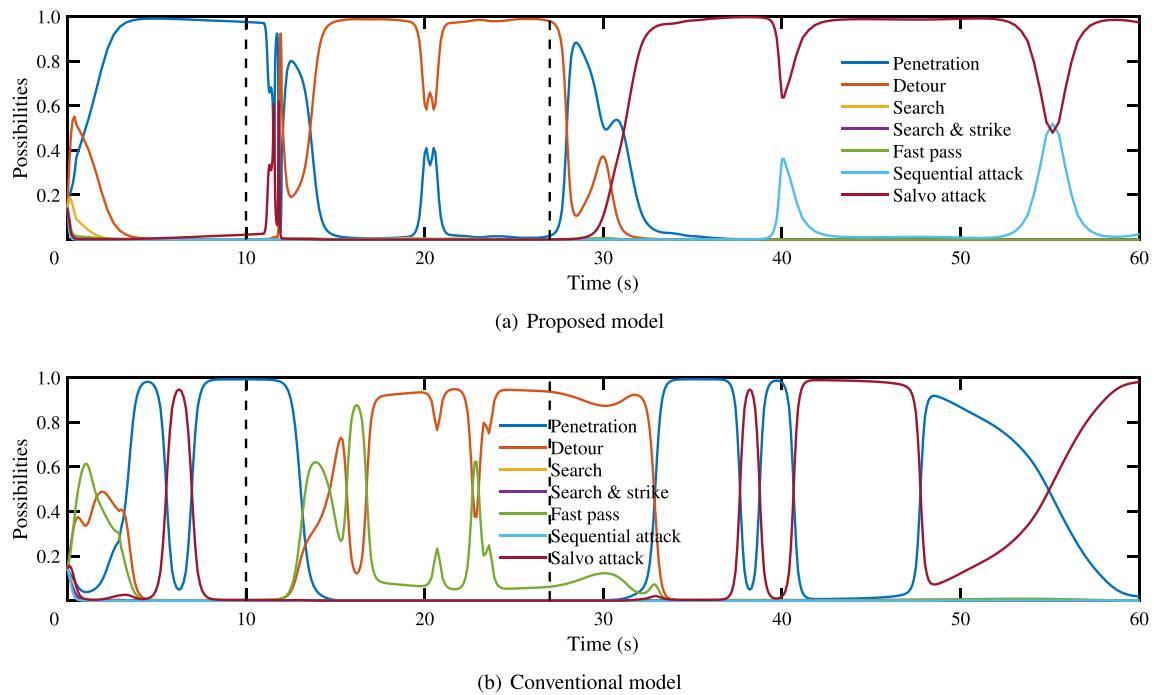


Fig. 18 Transition process of Scenario 2.

world issues, it is just a simplification of real-world noise. Flight experiments introduce various unpredictable noise, such as wind gusts, sensor inaccuracies or even slight calibration errors. The scaled-down operation also amplified measurement errors.

7. Conclusions

In this paper, we introduced the concept of DDP image and propose a swarm intention identification scheme. The DDP image provides an intuitive visualization of the noise distribution, which can be swiftly assimilated by the scheme. Our extensive numerical simulations have demonstrated the clear superiority of our proposed method over traditional networks. We have validated its robustness and accuracy in both simple and complex scenarios, showcasing its versatility in varying conditions. Additionally, real-world flight experiments have been conducted to further affirm the practical applicability and accuracy of our approach in intention identification. We have also delved into the robustness of our proposed model under varying image resolutions. Our findings indicate that higher resolution images significantly enhance the model's training success rate and its resilience to noise. This insight is crucial for balancing computational efficiency and accuracy in practical applications.

Our future research directions may include integrating intention identification with multi-objective tracking methods and recognizing swarm intentions in scenarios with incomplete or missing detection information.

CRedit authorship contribution statement

Yinhan WANG: Data curation, Formal analysis, Investigation, Methodology, Software, Validation, Visualization, Writing –

original draft, Writing – review & editing; **Jiang WANG:** Conceptualization, Data curation, Investigation, Resources, Validation, Writing – original draft, Writing – review & editing; **Shaoming HE:** Conceptualization, Funding acquisition, Investigation, Methodology, Project administration, Resources, Supervision, Writing – original draft, Writing – review & editing; **Fei WANG:** Conceptualization, Data curation, Validation, Visualization, Writing – original draft; **Qi WANG:** Conceptualization, Data curation, Validation, Writing – original draft.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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